Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

B.Sc. (Engg.) 3rd Year, 2nd Semester Final Examination-2024

Course Code: EEE 339 **Session: 2021-2022**

Course Title: Power Electronics

Total Marks: 100 **Credits: 3.0** **Total Time: 3 Hours**

[The figures in the right margin indicate full marks]

Part-A

*[Answer any **TWO** of the following questions]*

1. (a) With a two-transistor model explain the characteristics of a Silicon Controlled Rectifier (SCR). 7

   (b) Explain how natural commutation and forced commutation occur, giving one static power semiconductor device example for each. 6

   (c) The holding current of the thyristors in the single-phase full converter is $I_H = 500\ mA$ and the delay time is $t_d = 1.5\ \mu s$. The converter is supplied from a 120 V, 60 Hz supply and has a load of $L = 10\ mH$ and $R = 10\ \Omega$. The converter is operated with a delay angle of $\alpha = 30°$. Determine the minimum value of gate pulse width, t_g. 7

   (d) Explain why BJT, SCR cannot be operated in parallel, whereas MOSFET's can operate in parallel. 5

2. (a) A single-phase bridge rectifier that supplies a very high inductive load such as a DC motor. The turns ratio of the transformer are unity. The load is such that the motor draws a ripple-free armature current of I_a. Determine (i) the HF of input current and (ii) the input PF of the rectifier. 12

   (b) For a three-phase full-bridge controlled rectifier derive an expression for the DC output voltage as a function of the firing angle α by explaining its operation. (Draw the waveforms of gate pulses, input phase currents and output voltage.) 13

3. (a) Distinguish between DC-DC converters and switching-mode regulators. 6

   (b) What is secondary breakdown? Explain why secondary breakdown occur in BJT and Thyristors but not in MOSFETs. 7

   (c) A three-phase full-wave converter in Fig. 1 is operated from a three-phase Y-connected 208 V, 60 Hz supply and the load resistance is $R = 10\ \Omega$. If it is required to obtain an average output voltage of 50% of the maximum possible output voltage, calculate (i) the delay angle α, (ii) the rms and average output currents, (iii) the average and rms thyristor currents, (iv) the rectification efficiency, (v) the TUF, and (vi) the input PF. 12

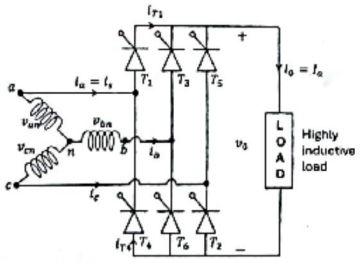


Fig. 1: Three phase full-wave converter for Q-3c

Part-B

*[Answer any **TWO** of the following questions]*

4. (a) Explain the working principle of a single-phase cycloconverter. 5

   (b) For the circuit of Fig. 2, $R = 2\ \Omega$, $L = 20\ mH$, and $V_{dc} = 100\ V$. The AC source is 120 V rms at 60 Hz. Determine (i) an expression for the current in the circuit, (ii) the power absorbed by the resistor, (iii) the power absorbed by the DC source, and (iv) the power supplied by the AC source and the power factor of the circuit. 10

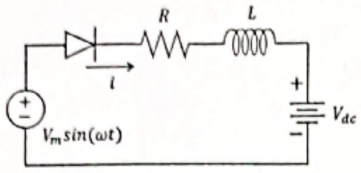


Fig. 2: Circuit for Q-4b

(c) The circuit of Fig. 3 has an AC source with $V_m = 100\ V$ at 60 Hz and a series RL load with $R = 10\ \Omega$ and $L = 10\ mH$. (i) Determine the average current in the load, (ii) Estimate the peak-to-peak variation in load current based on the first AC term in the Fourier series, (iii) Determine the power absorbed by the load and the power factor of the circuit and (iv) Determine the average and rms currents in the diodes. 10

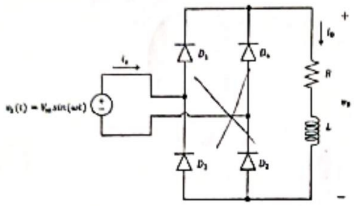


Fig. 3: Circuit for Q-4c

5. (a) What makes the SEPIC converter superior over buck-boost converter? Explain. 6

(b) The buck-boost circuit has these parameters: $V_s = 24\ V$, $D = 0.4$, $R = 5\ \Omega$, $L = 20\ \mu H$, $C = 80\ \mu F$, $f = 100\ kHz$. Determine the output voltage and average inductor current. 6

(c) Design a multilevel inverter circuit for a five level operation and explain the operation of the circuit. 5

(d) Describe the operation of a single-phase voltage controller of Fig. 4 with input voltage, output voltage, input current, output current for $\alpha = 90°$. Also, deduce an expression for output rms voltage. And find its value when input voltage is 120 V (rms). 8

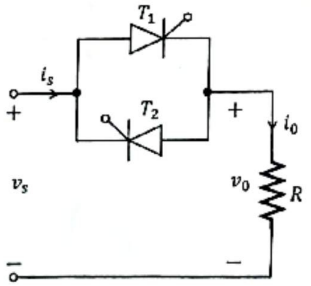


Fig. 4: Single phase voltage controller for Q-5d

6. (a) For a three-phase bidirectional AC voltage controller, draw the circuit diagram. Also draw the waveforms for input phase voltages, Thyristor gate pulses, and output phase voltage (for $\alpha = 60°$). 8

(b) What is the advantage of Cuk regulator over Buck-Boost regulator. 6

(c) A parallel resonant DC-DC converter has $V_S = 100\ V$, $Lr = 8\ \mu H$, $Cr = 0.32\ \mu F$, $R_L = 10\ \Omega$ and $f_s = 120\ kHz$. Determine the output voltage of the converter. Assume the output filter components L_o and C_o produce a ripple-free output current and voltage. 6

(d) Draw the schematic diagram of stand-alone solar PV system, describe the function of main components used in it. 5